

Few-Shot Prompt-Based Longitudinal Mood-State Analysis of Bipolar Disorder on Social Media

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Abstract. Bipolar disorder (BD) is frequently misdiagnosed as major depressive disorder, and monitoring mood-state *transitions* over time is critical for timely intervention. Existing social media datasets for BD research typically provide only binary diagnostic labels or per-post mood scores; few capture mood trajectories. Because expert annotation is costly and fine-tuning requires scarce labeled data, we propose a few-shot prompt-based LLM method that requires no task-specific training. The annotation schema is grounded in DSM-5 episode criteria and accompanied by eight synthetic few-shot examples, making the method directly applicable to other BD corpora. The method labels posts at two granularities: per-post mood state and 14-day period-level trends (dominant state, trend direction, change points), enabling trajectory modeling that per-post labels alone cannot support. **Using Gemini 3.1 Pro, we apply the method to 105 self-identified BD users on BD-focused subreddits (1,794 14-day periods, 15,423 posts and comments, April 2019–May 2026), observing depressive-pole predominance broadly consistent with clinical expectations for BD-related online discussion.** Post-level classification is externally validated against the BD-Risk dataset [1] on a held-out, author-disjoint, stratified subset of 145 posts, achieving macro F1 of 0.519 with high depressive recall (87.9%) and lower manic-pole recall (35.7%/6.7%). Six error patterns are characterized, including a structural label-text consistency issue at the manic pole that bounds achievable manic recall.

Keywords: Bipolar Disorder · Large Language Models · Few-Shot Prompting · Social Media · Clinical NLP · Mood Trajectory

1 Introduction

Bipolar disorder (BD) is characterized by recurrent episodes of mania, hypomania, and depression, affecting 1–2% of the global population [2]. An estimated 17–50% of BD cases are initially misdiagnosed as major depressive disorder (MDD), because patients typically seek help during depressive episodes and may not recognize manic or hypomanic states as pathological [3, 4]. This misdiagnosis leads to inappropriate treatment (e.g., antidepressant monotherapy, which may trigger manic switching) and delays proper intervention.

Social media platforms such as Reddit host BD communities (e.g., r/bipolar, r/BipolarReddit) where users openly discuss symptoms, treatment, and daily functioning. Prior work has used these data for BD and MDD classification [5–7], yet existing datasets provide only binary diagnosis labels (BD vs. MDD) or user-level risk scores. Few resources offer post-level mood state labels tracked over time, limiting computational research on mood trajectories and, in turn, on BD progression and early intervention.

Constructing expert-annotated BD corpora is costly: mood-state labeling requires psychiatric expertise, and the scarcity of labeled data makes fine-tuning approaches difficult to scale. At the same time, recent advances in large language models (LLMs) suggest that they can follow human-written guidelines with reasonable accuracy on text annotation tasks [8]. This motivates our approach: if LLMs can internalize clinical guidelines presented in a prompt, then a carefully designed DSM-5-grounded prompt with synthetic few-shot examples should enable mood-state annotation *without task-specific fine-tuning or labeled training data*, making the method directly applicable to new BD corpora.

Recent work has applied LLMs to mental health NLP tasks [9, 10]; however, whether they can classify BD mood states at expert-level accuracy, especially manic-pole states that are difficult to detect from text, has not been established.

This paper proposes a few-shot prompt-based LLM method for longitudinal mood-state analysis of BD on social media and demonstrates its application to a Reddit cohort. We collect Reddit posts from users who self-identify as having a BD diagnosis in BD-focused subreddits (r/bipolar, r/BipolarReddit, r/bipolar2) and annotate them at two granularities: (1) per-post mood state (*Depressive*, *Stable*, *Hypomanic*, *Manic*) and (2) 14-day period-level mood trends (dominant state, trend direction, change points). The annotation schema follows DSM-5 episode criteria and is implemented as an LLM pipeline (Gemini 3.1 Pro). We validate the post-level annotations against the BD-Risk dataset [1] on a held-out, author-disjoint, stratified subset of 145 posts, achieving macro F1 of 0.519 with 87.9% depressive recall and 35.7%/6.7% recall on hypomania/mania. The recall asymmetry across poles reflects both a known model property (manic-side states often manifest through described behaviors rather than affective tone) and a structural label-text consistency issue with the source dataset’s manic-pole labels.

Our contributions:

1. **Method:** A DSM-5-grounded few-shot prompt schema for LLM-based mood-state annotation at two temporal granularities (per-post state and 14-day period-level trend), requiring no fine-tuning or labeled training data.
2. **Evaluation:** External validation of post-level state classification against the BD-Risk expert-labeled dataset [1] on a held-out, author-disjoint, stratified subset, with macro F1 of 0.519 (87.9% depressive recall, 35.7% hypomanic recall, 6.7% manic recall), complemented by a zero-shot baseline comparison and a cross-model feasibility probe.
3. **Demonstration:** Application of the method to 105 self-identified BD users from BD-focused subreddits (1,794 14-day periods, 15,423 posts and comments

spanning April 2019 through May 2026), producing longitudinal mood trajectory annotations with distributions broadly consistent with clinical expectations for BD-related online discussion.

2 Related Work

2.1 Longitudinal Mood Monitoring in BD

Clinical efforts to track BD mood trajectories rely heavily on ecological momentary assessment (EMA) and digital phenotyping with smartphone sensors and self-report apps [11, 12]. These approaches yield dense, high-frequency mood signals; however, they require active patient enrollment and consent, limiting cohort size and external use. Social media offers a complementary unobtrusive source: it provides passive, naturally produced language data over months to years for users already discussing their condition. Publicly available social-media corpora with period-level mood trajectory annotations remain limited, which constrains method development for longitudinal BD analysis.

2.2 Social Media in Mental Health Detection

De Choudhury et al. [13] showed that social media signals can predict depression onset. The SMHD dataset [5] covers nine mental health conditions via self-reported diagnoses; Coppersmith et al. [6] established shared tasks for depression and PTSD detection from X (formerly Twitter).

For BD, Sekulić et al. [7] proposed Reddit-based classification and Jagfeld et al. [14] compiled a large BD Reddit corpus; however, both rely on self-reported diagnoses without expert validation [15, 16]. The BD-Risk dataset [1] provides per-post mood labels validated by a psychiatrist and a clinical psychologist; we use it as the gold standard for our post-level validation.

2.3 LLMs for Clinical NLP and Mental Health

Xu et al. [9] evaluated LLMs on mental health prediction from online text; Yang et al. [10] fine-tuned MentalLLaMA for interpretable mental health analysis. Lee et al. [1] found that ChatGPT achieved only an F1 of 0.130 on BD risk detection (vs. 0.578 for their multi-task model), showing that off-the-shelf LLMs struggle with BD-specific tasks. On the more general question of LLM annotation quality, Gilardi et al. [8] show that ChatGPT can match or exceed crowd workers on text annotation tasks; our work follows this line by treating the LLM as an annotator (not a classifier) and grounding its outputs in an explicit DSM-5-derived schema.

Our work differs from these evaluations in that we do not predict diagnosis. Instead, we propose a prompt-based method that *annotates* per-post mood states and period-level trends, and validate the annotations against expert labels. The error analysis (Section 4.5) characterizes the failure modes that the schema must address and that downstream users should remain aware of.

3 Methodology

3.1 Method Overview

We apply the proposed method to Reddit data from users who self-identify as having a BD diagnosis in BD-focused subreddits, producing a longitudinal mood-state-labeled corpus at two temporal granularities (per-post state and 14-day period-level trend). Fig. 1 shows the end-to-end flow from data collection through LLM-based annotation.

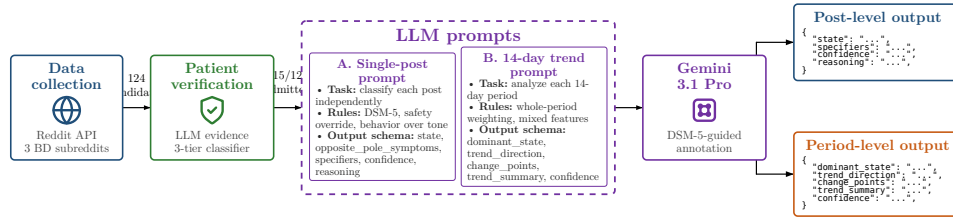


Fig. 1. Annotation pipeline: data collection (3 BD subreddits) → LLM patient verification (three-tier classifier) → Gemini 3.1 Pro annotation with two DSM-5-grounded prompts (single-post + 14-day trend) → structured JSON outputs at the two temporal granularities.

Data collection. We continuously crawl three BD-focused subreddits (r/bipolar, r/BipolarReddit, r/bipolar2) via the Reddit API, retrieving each active author’s full posting history (submissions and comments) with periodic re-crawls to capture ongoing activity.

Patient verification. Posting to a BD-focused subreddit is a necessary yet insufficient signal: many such posts come from clinicians, family members, or community discussion. To screen the candidate pool, we apply an LLM three-tier classifier (Gemini 3.1 Pro, separate prompt) that scans each author’s full posting history and returns **verified** (explicit first-person diagnostic statements, e.g., “I was diagnosed with bipolar II in 2019”, or specific treatment/hospitalization narratives), **probable** (consistent self-identification through symptoms, medication, or community-membership tone without an explicit diagnosis statement), or **unverified** (no diagnostic signal). Only the **verified** and **probable** tiers are admitted to the annotation cohort; this matches the inclusion model used in prior BD social-media datasets [7, 14] with stricter per-user evidence gating than a one-post membership rule.

Scale. The verified+probable cohort comprises 115 of 124 candidate authors. Users with fewer than two 14-day periods containing posts (i.e., insufficient longitudinal span for trend analysis) are excluded. The remaining 105 users contribute 2,611 submissions and 12,812 comments spanning April 2019 through May 2026, yielding 1,794 valid analysis periods.

3.2 Annotation Schema

Our annotation schema draws on DSM-5 episode definitions [17], operationalizing them as a structured prompting framework for LLM-based annotation. The rules below were developed iteratively against an error analysis on external expert labels (see Section 4.5); each clinical-guidance rule named below is the schema’s response to a recurring failure mode characterized there. Because the BD-Risk data-use agreement prohibits verbatim reproduction of post text, the error analysis illustrates each pattern with a synthetic case that preserves the clinically relevant structure of the original disagreement.

Post-Level State Classification

For each individual post (submission or comment), the LLM assigns a categorical mood state from five options:

- **Manic:** Grandiosity, pressured writing (run-on sentences, excessive capitalization), flight of ideas (tangential topic shifts), extreme irritability or euphoria.
- **Hypomanic:** Elevated energy and pace with maintained coherence, social disinhibition, uncharacteristic intensity without psychotic features.
- **Depressive:** Linguistic constriction, absolutist language (“never,” “nothing”), high self-focus (first-person pronouns), cognitive distortions, suicidal ideation.
- **Stable:** Balanced emotional tone, metacognitive reflection, proportionate responses, community-supportive language.
- **Uncertain:** Reserved for truly uninterpretable posts; the LLM must attempt classification before resorting to this label.

The framework also supports a `with_mixed_features` specifier (following DSM-5 mixed-features criteria). Before applying this specifier, the prompt requires the LLM to extract an explicit list of opposite-pole symptoms, and only assigns `with_mixed_features` when three or more clear opposite-pole symptoms are documented. This evidence-extraction step prevents the mixed-features specifier from being used as a vague neutral label.

Period-Level Trend Analysis

For longitudinal mood trajectory modeling, we partition each user’s posting history into consecutive fixed-length periods (default: 14 days). The 14-day window length was set in consultation with clinical advisors and is grounded in DSM-5 episode-duration criteria, which define a major depressive episode as lasting at least two weeks and a manic episode as lasting at least one week [17]; a 14-day window therefore captures the minimum duration of a full depressive episode and allows observation of manic-episode onset and progression. Periods are anchored at the user’s first post (day 0) and advance in strict half-open intervals $[t_k, t_k + 14)$; submissions and comments are jointly assigned to the period containing their timestamp. All periods from the user’s first to last post are defined; periods without posts receive a `NO_DATA` label rather than being skipped,

preserving a continuous time grid for trajectory modeling. Fig. 2 illustrates the segmentation.

Period 1 day 0–13	Period 2 day 14–27	Period 3 day 28–41	Period 4 day 42–55	Period 5 day 56–69	Period 6 day 70–83
● ○ ● ○	○ ○	NO_DATA	● ○	○ ○ ○ ○	●

Fig. 2. Period segmentation (illustrative): fixed 14-day windows anchored at the user’s first post (day 0). Submissions (filled circles) and comments (open circles) fall in the period containing their timestamp; empty periods (Period 3) carry the `NO_DATA` label rather than being skipped.

For each period containing at least one post, the LLM analyzes the collected posts and produces:

- **Dominant state:** The primary mood state across the period (same five-class set as post-level), aggregated across the posts in the window.
- **Trend direction:** `NO_TREND` (state maintained), `TOWARDS_MANIA` / `TOWARDS_DEPRESSION` (progressive worsening toward the respective pole), or `FLUCTUATING` (alternation without a clear direction).
- **Change points:** Specific dates or events where a mood shift occurred, with pre- and post-shift states documented.
- **Trend summary:** A concise narrative describing the period’s trajectory and the evidence supporting the dominant state.
- **DSM-5 specifiers:** `with_mixed_features` when opposite-pole symptoms co-occur within the period (distinguished from sequential fluctuation).

Together, the two granularities support both event-level analysis (e.g., what preceded a state change) and longitudinal trajectory modeling, with the explicit change-point fields enabling change-point detection [18] at the textual level.

Few-Shot Example Construction

The post-level prompt is paired with eight synthetic few-shot examples (labeled A–H), written by the authors and never drawn from BD-Risk. Each exercises a schema rule targeting a failure mode observed during development (full analysis in Section 4.5): A demonstrates *Behavior Over Tone* on retrospective manic-side narration; B and E demonstrate the *SAFETY OVERRIDE* rule with its grandiose-mania exception; C and D contrast *Improvement-Narrative* against *Whole-Post Evidence Weighting*; F counters the default-to-*Uncertain* tendency on short posts; G demonstrates the *Recurrent-Pattern Exception* for substance-triggered hypomania; H exercises *Severity Descriptors* for *Hypomanic-vs-Stable* boundaries. Each example provides the input text with the full expected JSON output (including the `opposite_pole_symptoms` evidence list and reasoning), so

the model observes both the target label and the evidence chain. The full texts are reproduced in the Appendix.

3.3 LLM Configuration

We use Gemini 3.1 Pro [19] through the official API with structured JSON output; the model was selected for its large context window and native structured-output generation. Each post is processed independently with the full annotation schema and the few-shot examples described above as the system instruction; the model returns a JSON object with `state`, `opposite_pole_symptoms`, `specifiers`, `confidence` (High/Medium/Low), and `reasoning` fields, where `opposite_pole_symptoms` carries the explicit evidence list required before `with_mixed_features` can be assigned (see Section 3.2). For period-level annotation the LLM additionally returns `trend_direction`, `change_points`, and a `trend_summary` narrative, with `confidence` on a 0–1 scale. We use the Gemini 3.1 Pro default temperature of 1.0; the model is not fine-tuned.

3.4 External Validation Against BD-Risk

The BD-Risk Dataset

The BD-Risk dataset [1] comprises 7,346 Reddit posts from 1,025 users, each carrying a psychiatrist-guided mood level label on a 7-point scale (−3 to +3). Because the dataset recruits users via an initial MDD presentation (MDD-only and MDD→BD groups), it is structurally enriched for depressive-pole content (88.9% of posts ≤ 0).

Gold State Derivation

The BD-Risk dataset provides only ordinal mood labels; categorical states are not directly annotated. To obtain gold states for evaluation, we derive them from BD-Risk mood labels using the mapping shown in Table 1.

Table 1. Mapping from BD-Risk 7-point mood labels to derived gold states. This mapping treats +1 as within the normal positive range (Stable) rather than pathological activation.

BD-Risk mood label	Derived gold state	Notes
−3, −2, −1	Depressive	
0, +1	Stable	+1 = high motivation / positive mood within normal range
+2	Hypomanic	Clear manic-side activation without psychosis
+3	Manic	Severe manic expression with psychotic features

Evaluation Set Construction

The full BD-Risk dataset exhibits a heavily skewed mood distribution (88.9% of posts ≤ 0). Because the deployment task is BD risk detection rather than general mood classification, we deliberately oversample manic-pole posts so that per-class metrics on the underrepresented classes are computed with sufficient support.

We separate the labeled posts into two disjoint subsets. A *development* subset (314 posts) is used during prompt design and failure-mode analysis. A *held-out* subset (145 posts) is used exclusively for the evaluation reported below; it is *author-disjoint* from the development subset, drawn by stratified sampling from BD-Risk authors not present in the development subset, with quotas ensuring sufficient per-class support across the four derived gold states (60 Depressive, 40 Stable, 30 Hypomanic, 15 Manic). All metrics in Section 4.1 are computed on the held-out subset; the development subset is never used to produce reported numbers. Manic-pole gold posts in BD-Risk come almost exclusively from the MDD $\rightarrow$ BD group, so the held-out manic-side samples are structurally MDD $\rightarrow$ BD-derived (a limitation discussed in Section 6).

3.5 Evaluation Metrics

Throughout this paper, we refer to the expert-assigned labels from the BD-Risk dataset as gold labels and the LLM outputs as predictions. Gold states are derived from BD-Risk mood labels via the mapping in Table 1.

We report per-class precision, recall, and F1, along with overall accuracy and macro F1, plus the confusion matrix. Two accuracy variants are reported: accuracy *excluding* Uncertain treats Uncertain outputs as abstentions and removes them from both numerator and denominator; accuracy *including* Uncertain counts Uncertain as incorrect, providing a conservative lower bound. Per-class precision, recall, and F1 are computed on the excluding-Uncertain basis. All metrics are computed on the full evaluation set.

3.6 Evaluation Design

Beyond the main BD-Risk holdout validation, we conduct two additional evaluations to characterize the schema’s properties:

- **Zero-shot baseline comparison:** To quantify how much the structured annotation schema (DSM-5 rules, few-shot examples) contributes beyond the LLM’s base capability, we re-evaluate the same model with a minimal zero-shot prompt containing only the task definition and output format.
- **Cross-model feasibility probe:** To test whether the schema generalizes beyond a single LLM provider, we additionally evaluate the full schema with OpenAI’s GPT-5.5, characterizing schema portability and the impact of provider-level content-policy differences on annotation feasibility.

4 Results

4.1 Post-Level Validation Against BD-Risk

We first validate the post-level state classification against BD-Risk expert labels (Table 2: per-class metrics with macro-aggregated summary).

Table 2. Per-class metrics with macro summary on the held-out subset (n=145; 138 excluding 7 Uncertain). Accuracy excl./incl. Uncertain = 65.9 % / 62.8 %; macro F1 is the primary metric because the subset is intentionally stratified.

State	Precision	Recall	F1	Support
Depressive	0.630	0.879	0.734	58
Stable	0.659	0.784	0.716	37
Hypomaniac	0.833	0.357	0.500	28
Manic	1.000	0.067	0.125	15
<i>Macro avg</i>	<i>0.781</i>	<i>0.522</i>	<i>0.519</i>	<i>138</i>

Depressive recall is high (87.9%) and Stable recall is moderate (78.4%), while Hypomaniac and Manic recall remain low (35.7% and 6.7%), indicating that the LLM correctly recognizes most depressive and stable posts while missing a majority of manic-pole cases. The confusion matrix (Table 3) makes the dominant error flow explicit: among the 30 gold-Hypomaniac posts, 12 are predicted as Depressive and 6 as Stable; among the 15 gold-Manic posts, 11 are predicted as Depressive and 2 as Stable. The Manic-to-Depressive error flow is a recurring pattern with a likely label-text origin that we discuss in Section 6.

Table 3. State confusion matrix on the held-out subset (rows: derived gold state; columns: LLM prediction). DEP = Depressive, STA = Stable, HYP = Hypomaniac, MAN = Manic, UNC = Uncertain.

Gold	Predicted					Total
	DEP	STA	HYP	MAN	UNC	
Depressive	51	7	0	0	2	60
Stable	7	29	1	0	3	40
Hypomaniac	12	6	10	0	2	30
Manic	11	2	1	1	0	15

These *Hypomaniac* errors are concentrated on gold +2 posts whose manic-side activation is masked by negative tone, a pattern that the current prompt does not fully resolve.

4.2 The Uncertain Label as Quality Control

The LLM assigned *Uncertain* to 7 posts (4.8%), abstaining when post content was insufficient for state assessment, in line with the prompt’s explicit instruction to prefer abstention over forced classification. *Uncertain* emissions are distributed across gold classes (2 *Depressive*, 3 *Stable*, 2 *Hypomanic*, 0 *Manic*), with no strong concentration on a single pole.

4.3 Schema Contribution: Comparison with a Zero-Shot Baseline

Following the design in Section 3.6, we compare the full schema against a minimal zero-shot prompt on the held-out subset. The zero-shot prompt retains only the task definition (post $\rightarrow$ one of five states) and the output JSON fields; all DSM-5 rules, *SAFETY OVERRIDE*, *Severity Descriptors*, and few-shot examples are removed. Table 4 reports the side-by-side metrics.

Table 4. Schema contribution on the held-out subset (n=145). Both runs use Gemini 3.1 Pro with the same JSON output; the only variable is the system prompt (zero-shot: task definition only; full schema: *SAFETY OVERRIDE*, *Severity Descriptors*, *Clinical Guidance*, eight few-shot examples).

Metric	Zero-shot	Full schema	Δ
Accuracy (incl. Uncertain)	51.7%	62.8%	+11.1 pp
Accuracy (excl. Uncertain)	63.6%	65.9%	+2.3 pp
Macro F1	0.459	0.519	+0.060
Uncertain count	27	7	−20
DEPRESSIVE F1	0.738	0.734	−0.004
STABLE F1	0.600	0.716	+0.116
HYPOMANIC F1	0.500	0.500	± 0.000
MANIC F1	0.000	0.125	+0.125

Three observations frame the schema’s value. First, the largest single effect is a $4\times$ reduction in *Uncertain* emissions ($27 \rightarrow 7$): the structured schema gives the LLM the vocabulary to commit to a label rather than abstain, which explains why the accuracy gain including *Uncertain* (+11.1 pp) is much larger than the gain excluding *Uncertain* (+2.3 pp). Second, *Stable* classification benefits the most in per-class F1 (+0.116), driven by the Severity Descriptors’ explicit “Stable includes mild positive activation” rule that prevents the LLM from defaulting non-pathological positive posts to *Depressive* or *Uncertain*. Third, *Depressive* and *Hypomanic* F1 are unchanged, suggesting that the full schema did not materially alter performance for those categories on this subset, and *Manic* remains poorly recalled in both runs (0/15 zero-shot vs. 1/15 schema, counting *Uncertain* as incorrect), further suggesting that the manic-pole limitation is structural (label-text consistency, Section 6) rather than a limitation resolved by this richer prompt.

The comparison suggests that the schema’s primary contribution is *coverage and decisiveness* (preventing abstention, anchoring border-class decisions) rather than raw classification accuracy on the cases the LLM is already confident about.

4.4 Cross-Model Annotation Feasibility

As a cross-model probe (Section 3.6), we evaluated the full schema with OpenAI’s GPT-5.5 (`reasoning_effort` = "high") on the same 145 held-out posts. The schema and JSON output format were identical to the main run; only the underlying model changed.

Refusal under content policy. GPT-5.5 declined to classify 103 of 145 held-out posts (71.0%) with a verbatim refusal ("I'm sorry, but I cannot assist with that request."), concentrated on posts containing explicit self-harm or suicidal content, the safety-relevant subset that BD-Risk includes by clinical design and that the *SAFETY OVERRIDE* rule is built to handle. The prompt’s clinical-research framing did not overcome the refusal. Gemini 3.1 Pro produced structured output for all 145 posts under the same prompt.

Performance on the non-refused subset. On the 42 posts GPT-5.5 did classify, both models score higher than on the full 145 because the subset is skewed away from depressive-crisis content (16 *Depressive*, 10 *Stable*, 12 *Hypomaniac*, 4 *Manic*). Table 5 reports the head-to-head metrics.

Table 5. Cross-model comparison on the 42 non-refused posts (of 145 held-out). The subset selectively excludes depressive-crisis posts refused by GPT-5.5 (103 posts, 71%). GPT-5.5 reaches higher macro F1, driven mostly by Manic F1 (2/4 vs. Gemini 1/4).

Metric (on 42 non-refused posts)	Gemini	GPT-5.5
Accuracy (excl. Uncertain)	72.5%	73.2%
Macro F1	0.649	0.710
Macro Precision	0.827	0.838
Macro Recall	0.656	0.679
DEPRESSIVE F1	0.800	0.769
STABLE F1	0.727	0.737
HYPOMANIC F1	0.667	0.667
MANIC F1	0.400	0.667
Uncertain emissions	2	1

Interpretation. When GPT-5.5 does respond, the schema produces sensible structured output, suggesting it is not inherently Gemini-specific. The head-to-head numbers favor GPT-5.5 on the non-refused subset, yet the subset selection itself is the dominant effect: GPT-5.5 systematically filtered out the hard depressive-crisis posts that drive most of Gemini’s error rate on the full held-out subset, so the macro-F1 comparison overstates GPT-5.5’s effective competence on a clinical mental-health corpus. Most importantly, in the configuration tested here, production safety filters can render an LLM *unsuitable as an annotator for*

psychiatric corpora: a 71% refusal rate makes GPT-5.5 infeasible as a stand-alone annotator regardless of intrinsic capability.

4.5 Error Analysis

We characterize six failure modes identified through manual analysis of LLM-vs-BD-Risk disagreements on the development subset. For each pattern, we name the schema rule designed to mitigate it and note whether the rule resolves the pattern on the held-out subset or whether it remains a residual error. Manic-pole posts misclassified as depressive or stable remain the dominant residual failure mode (quantified in Section 4.1).

Pattern 1: Retrospective behavioral cues ignored (dominant manic-side error). When users describe manic-episode behaviors (impulsive spending, aggressive confrontations, hyperactivity) in a retrospective post written with remorse or self-blame, the LLM anchors on the *current emotional tone* rather than the *clinical significance of the described behaviors*, and predicts *Depressive*. The schema’s *Behavior Over Tone* rule directly targets this conflation; however, it remains the dominant residual error on the held-out subset, indicating that the rule reduces the pattern without fully eliminating it. *Synthetic case*: a user recounts a week of reckless spending and impulsive decisions in a tone of deep regret and self-condemnation; gold is *Hypomanic* (the behaviors are hallmark manic symptoms), the LLM predicts *Depressive* (anchored on the self-deprecating tone).

Pattern 2: Mixed features conflated with neutrality. When a post carries symptoms from both poles simultaneously (e.g., severe sleep disruption and inability to concentrate alongside aggressive outbursts), the LLM sometimes treats the coexistence as cancellation and defaults to *Stable*. Under DSM-5 [17], mixed features should instead surface as a `with_mixed_features` specifier on the dominant pole. The schema’s mixed-features rules reduce this conflation, though residual instances remain when symptom signals are subtle. *Synthetic case*: a user writes that they have not slept in three days, cannot focus at work, and snapped at their partner, all in the same post; gold is *Hypomanic* with mixed features (decreased need for sleep alongside dysphoric irritability), the LLM predicts *Stable* on the reasoning that the signals “balance out”.

Pattern 3: Calm writing style masking suicidal ideation (safety-critical). Some posts express suicidal ideation (“I want to die”) in calm, reflective, or educational prose; the LLM reads the linguistic register as stable and classifies accordingly, while the gold state is *Depressive* (−3). Calm writing does not rule out suicidal crisis, and the schema’s explicit *SAFETY OVERRIDE* rule forces *Depressive* whenever crisis-level language is present regardless of surrounding tone. On the held-out subset, no *Stable* or *Uncertain* predictions were observed for posts with explicit suicidal content, suggesting that the rule addresses this pattern on the evaluated subset. *Synthetic case*: a user posts a coherent essay about public misconceptions of depression; embedded in the second paragraph, in

the same calm register, is a single sentence noting that the writer quietly thinks about not waking up most mornings.

Pattern 4: End-of-post hope overriding pervasive impairment. Posts describing severe functional impairment (academic collapse, inability to maintain daily routines, social withdrawal) sometimes close with a single hopeful sentence (e.g., “my therapist said maybe I shouldn’t give up”), and the LLM anchors on this terminal positive signal (consistent with a recency bias) to predict *Stable* while the gold label tracks the pervasive impairment throughout. The schema’s *Whole-Post Evidence Weighting* rule addresses this by instructing the LLM to weigh the dominant clinical picture over terminal sentiment; the zero-shot comparison (Table 4) shows that the full schema improves *Stable* F1 by +0.116, consistent with reduced over-assignment of *Stable* to impaired posts. *Synthetic case:* a user reports missing every class for a month, eating almost nothing this week, and losing the ability to reply to friends, closing with “maybe tomorrow will be different”; gold is *Depressive* on the pervasive impairment, the LLM predicts *Stable*.

Pattern 5: Substance-induced activation vs. endogenous mood. When users describe mood elevation explicitly attributed to substances (e.g., caffeine-induced euphoria described as feeling “on top of the world”), the LLM may interpret these cues as hypomanic mood, while gold labels distinguish acute pharmacological activation from endogenous baseline. The schema’s *Substance vs. Endogenous Mood* rule addresses this boundary and also covers the *Recurrent-Pattern Exception* where an unusual reactivity to a common substance is itself bipolar-spectrum. This pattern is rare in the held-out subset; the rule’s primary effect is preventing false *Hypomanic* predictions on substance-attributed posts. *Synthetic case:* a user writes that after a fourth coffee they suddenly feel “unbeatable and ready to overhaul” their apartment, attributing the surge to caffeine and expecting an evening crash; gold is *Stable* (acute, externally caused, self-labeled), the LLM predicts *Hypomanic* on the surface cues.

Pattern 6: Uncertain masking embedded clinical signals. Uncertain functions appropriately in the large majority of cases; the failure described here is rare. The LLM correctly identifies the post as metacommentary or informational, then fails to surface clinically significant content embedded inside. *Synthetic case:* a user writes a meta-commentary criticizing how recovery is portrayed online and, mid-argument, notes parenthetically that they have been thinking about ending things; gold is *Depressive*, the LLM defaults to *Uncertain* on the meta-discussion register. The *SAFETY OVERRIDE* rule mitigates this pattern by forcing *Depressive* on crisis-level language regardless of overall framing. Like Pattern 3, the rule appears effective on the held-out subset: no crisis-level posts were classified as *Uncertain*.

5 Longitudinal Demonstration: Period-Level Mood Trends

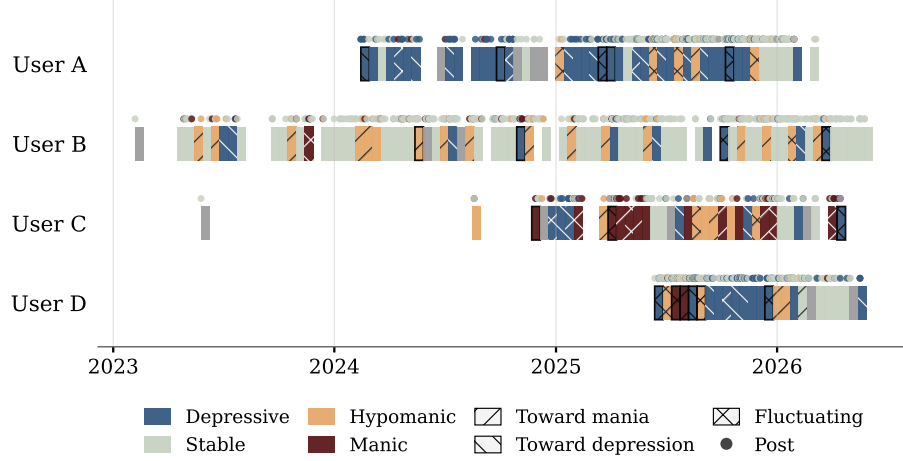


Fig. 3. Mood trajectories for four anonymized users (A: depressive with fluctuation, 49 periods; B: hypomanic-leaning with frequent manic transitions, 74 periods; C: manic-dominant with rapid cycling, 36 periods; D: dense posting with high mixed-features incidence, 25 periods). Bars = 14-day periods (color = dominant state; hatch = trend direction; thin black border = `with_mixed_features`); dots above bars = post-level annotations, color-coded by post state.

Table 6. Period-level distributions on the 105-user cohort ($n = 1,794$). Left: trend directions (NO_TREND = state maintained, FLUCTUATING = alternation, TOWARDS_MANIA / TOWARDS_DEPRESSION = worsening). Right: dominant states; `with_mixed_features` was applied to 84 periods (4.7%).

Trend direction	Periods	%	Dominant state	Periods	%
NO_TREND	1,506	83.9%	Stable	765	42.6%
FLUCTUATING	106	5.9%	Depressive	564	31.4%
TOWARDS_DEPRESSION	105	5.9%	Hypomanic	157	8.8%
TOWARDS_MANIA	77	4.3%	Manic	54	3.0%
			Uncertain	254	14.2%

Most 14-day windows show NO_TREND, as expected: DSM-5 episode-duration criteria require ≥ 1 week for mania, ≥ 4 days for hypomania, and ≥ 2 weeks for major depression [17], and untreated episodes typically last weeks to months, so within-window transitions are infrequent; the rare TOWARDS_DEPRESSION and TOWARDS_MANIA trends mark episode onset or escalation, the signals most relevant for early intervention. Stable and Depressive states dominate (42.6% and 31.4% of periods), while manic-pole states constitute a smaller share (Hypomanic 8.8%, Manic 3.0%), broadly consistent with the depressive-pole predominance reported in long-term BD cohort studies [2] while reflecting the selection and posting biases of Reddit communities.

Post-level state distribution. Table 7 breaks the assignments down by content type; the submission/comment asymmetry and its implications are discussed in Section 6. The *Uncertain* rate is low for both content types ($\leq 3.0\%$), reflecting the model’s confidence on the corpus.

Table 7. Post-level state distribution across the 105-user cohort, separated by content type. Posts (submissions, $n = 2,611$) function as longer-form emotional disclosures and carry the majority of polar-state labels; comments ($n = 12,812$) are predominantly conversational replies labeled Stable.

State	Posts	%	Comments	%
Manic	98	3.8%	93	0.7%
Hypomanic	277	10.6%	292	2.3%
Stable	1,217	46.6%	11,287	88.1%
Depressive	961	36.8%	755	5.9%
Uncertain	58	2.2%	385	3.0%

6 Discussion

Period-level trends. Prior BD datasets primarily provide per-user labels [5, 7] or per-post scores [1], leaving period-level mood trajectories underrepresented; our period-level annotations add this dimension by recording in each 14-day window the dominant state, whether mood is shifting (trend direction), and when shifts occur (change points). From the trend distribution (Section 5), four observations stand out. (1) TOWARDS_MANIA and TOWARDS_DEPRESSION trends are rare yet among the most clinically significant signals, as they mark episode onset where intervention has the most impact. (2) FLUCTUATING periods may correspond to rapid cycling or mixed presentations that single-post labels cannot capture. (3) Post-level states and period-level trends together enable hierarchical modeling: predicting the next period’s trajectory from the sequence of post-level features. (4) The dominant-state distribution preserves meaningful manic-pole representation, in contrast to MDD-recruited cohorts (BD-Risk: 88.9% depressive-

or-neutral); this balance matters for downstream models that must distinguish manic-pole from depressive states. The trend distributions are broadly consistent with clinical expectations for BD-related online discussion; direct external validation would require period-level expert annotations. A complementary direction is to model the absence signal in **NO_DATA** periods: reduced posting is sometimes associated with depression in digital-phenotyping research [11], yet the relationship is heterogeneous, so absence modeling would require posting-frequency baselines beyond a text-based pipeline.

Submission–comment asymmetry. Table 7 shows submissions and comments differ sharply: 51.2% of submissions carry a polar state vs. 9.1% of comments, which are dominated by *Stable* (88.1%). Submissions are longer-form disclosures while comments are short replies. Two downstream implications: a per-post classifier on a comment-heavy corpus is likely to appear over-confident on *Stable*, so per-content-type metrics are preferable to a single aggregate; and trajectory models should either up-weight submissions or rely on the period-level dominant-state annotation (which already aggregates across content types within the window) as the primary trajectory signal.

Manic-pole underdetection. The LLM achieves 87.9% recall for *Depressive* yet only 35.7% for *Hypomanic* and 6.7% for *Manic* on the held-out subset. Two factors compound this asymmetry. First, a model property: depressive language has stereotypical surface markers (negativity, self-focus, hopelessness), while manic-pole states often manifest through *described behaviors* (spending sprees, reduced sleep need, grandiose plans) narrated in any tone, and the LLM reads tone rather than the clinical significance of the behaviors described (see Pattern 1 in Section 4.5). Second, a label–text consistency issue in the BD-Risk annotation rule: as specified by [1] (Section 3.2), “posts exhibiting both manic and depressive moods are regarded as manic moods”, an asymmetric tie-breaker that elevates any mixed manic+depressive post to the manic side. Combined with the dataset’s MDD→BD recruitment, this yields gold-label Manic (+3) posts whose textual content matches BD-Risk’s own definition of −3 (“extreme anxiety and having suicidal thoughts”). A single-post LLM applying clinical-safety priors (classifying explicit self-harm content as *Depressive*) will systematically underperform on these posts, since the only signal it has is the very content the labeling rule overrode. The Manic-recall ceiling of 6.7% in Table 2 thus reflects a structural mismatch, not solely a model limitation (downstream implications in Limitations).

7 Limitations

The proposed method and corpus are subject to constraints that we group into evaluation scope, evaluation labels, manic-pole interpretability, cohort framing, and release-time reliability.

Evaluation scope. The main BD-Risk validation reports a single annotator (Gemini 3.1 Pro); a complementary cross-model probe with GPT-5.5 (see Table 5) suggests that the schema is not inherently Gemini-specific, yet GPT-5.5 cannot

be used at scale under OpenAI’s current content policy (71% refusal on the held-out subset), so the per-class numbers on the 145-post held-out subset should be read as Gemini-specific. The quantitative validation is also at the post level only, because few per-post expert-labeled BD datasets at sufficient scale are available for research use (BD-Risk was obtained through a formal data request and ethics review, and comparable shared-task corpora face similar access barriers); period-level trends, a key contribution of the method, are not externally validated.

Gold-state derivation. Most importantly, gold states are derived from BD-Risk mood labels via a deterministic mapping (Table 1) rather than being directly annotated by experts as categorical states. This mapping introduces imprecision: the boundary between adjacent categories is inherently uncertain (e.g., BD-Risk mood label +1 may reflect mild hypomania rather than stable mood), and the ordinal intensity score does not always correspond to a categorical clinical judgment (e.g., a recovery narrative within an ongoing depressive episode may receive a neutral mood score while remaining clinically depressive). Inter-expert agreement (Krippendorff’s $\alpha = 0.87$) is measured at the ordinal level, and mapping to categorical states amplifies disagreement at boundaries. Consequently, some apparent misclassifications may reflect mapping artifacts, and the reported accuracy figures should be interpreted as lower bounds on the schema’s true reliability.

Manic-pole interpretability. Two factors limit how the Manic-pole numbers can be read. First, the held-out Manic class is small ($n = 15$, because the entire BD-Risk dataset contains only 28 mood-+3 posts); the 95% confidence interval around the Manic-F1 estimate is wide ($\pm$ approximately 13 percentage points), and small Manic-F1 differences should not be interpreted as significant. Second, the Manic-recall ceiling of 6.7% is bounded by a structural mismatch between BD-Risk’s asymmetric tie-breaking rule and the textual evidence available to a per-post classifier (full analysis in Section 6); recovering label-text consistency would require either re-annotation under text-only criteria or a longitudinal evaluation protocol that supplies the temporal context the annotators had access to.

Cohort framing. Patient verification is LLM-based and screens for self-disclosure of a BD diagnosis in the user’s posting history; it does not constitute clinical confirmation. This follows the inclusion model used in prior BD social-media datasets [7, 14] with stricter per-user evidence gating than a one-post membership rule. Some users in the **verified** or **probable** tiers may describe a BD diagnosis without one having been clinically established, and the cohort conversely excludes users with BD who never disclose the diagnosis publicly. Reddit is also a self-selected, asynchronous channel, so the relationship between its posts and clinically observed mood states requires further investigation. Downstream uses that require clinician-confirmed BD status should treat the cohort as an LLM-screened self-identified sample rather than a clinical cohort.

Release-time reliability. All period-level (1,794) and post-level (15,423) annotations are produced entirely by the LLM; we have not yet conducted manual spot-checking on a stratified sample across mood states, confidence levels, and

trend directions. The held-out validation suggests stronger reliability for *Depressive* labels (87.9% recall, 73.4% F1) than for manic-pole labels; *Hypomanic* labels have moderate reliability (35.7% recall, 50.0% F1) with high precision (83.3%) when the label is assigned; *Manic* labels should be used with caution given both the small support and the manic-pole interpretability constraints above.

8 Conclusion

The validation separates the proposed method’s two failure sources: 87.9% depressive recall suggests that the schema captures dominant depressive-pole markers with reasonable coverage, while the 6.7% manic recall appears strongly influenced by a structural label-text consistency issue at the manic pole rather than by prompting alone (see Section 6). Because no external longitudinal ground truth exists at the period level, the corpus’s 1,794 trajectories are validated only indirectly through their post-level constituents. Three near-term priorities follow: (1) expert annotation of period-level trends to enable direct longitudinal validation; (2) a stratified human-in-the-loop audit across mood states, confidence levels, and trend directions, reported as inter-annotator agreement; and (3) extending the cross-model probe (Anthropic Claude, open-weight reasoning models) beyond the single-vendor evaluation, currently constrained by the production-safety refusal pattern documented for GPT-5.5. The resulting corpus is intended for computational mental health research and should not be used as a clinical diagnostic tool.

9 Ethical Considerations

This study involves analysis of publicly posted social media content discussing sensitive mental health experiences. The study protocol was reviewed and approved by the Research Ethics Committee of the University of Tsukuba (approval no. 25-188). We additionally adhere to Reddit’s privacy policy and social media research ethics guidelines [15].

De-identification. Before public release, all post content undergoes LLM-based de-identification. We classify personally identifiable information into five risk-ranked categories: *identifiers* (direct names), *quasi-identifiers* (combinable details such as specific locations, organizations, dates, or unique personal circumstances), *contact information*, *linkage codes*, and *personal identification codes*, replacing each detected PII span with a category-specific placeholder (e.g., [IDENTIFIER], [QUASI_ID]). The LLM additionally surfaces long-span quasi-identifiers where the accumulation of individually innocuous details (occupation, location, family structure) could jointly narrow identification to one person, a pattern rule-based or NER approaches typically miss. The taxonomy is designed to preserve clinically relevant content (medication names, diagnosis types, symptom descriptions, relative temporal expressions) while removing identifying information.

Data minimization and intended use. The dataset retains only text content and temporal information necessary for mood-state annotation; author usernames are replaced with anonymized identifiers, and subreddit membership and post metadata that could facilitate re-identification are excluded from the published dataset. To reduce search-based re-identification risk, released timelines use relative temporal offsets rather than exact posting times. Because LLM-based de-identification may miss residual quasi-identifiers, public release is conditioned on a pre-release privacy audit of a stratified sample. The dataset is intended solely for computational mental-health research and must not be used for re-identification, commercial profiling, or clinical decision-making without appropriate expert oversight.

A Appendix: Annotation Prompts

Three system prompts drive the pipeline: (A) the post-level prompt with the full DSM-5-grounded schema, clinical guidance rules (*SAFETY OVERRIDE*, *Behavior Over Tone*, *Whole-Post Evidence Weighting*, *Improvement-Narrative*, *Substance vs. Endogenous Mood*), mixed-features rules, and eight synthetic few-shot examples; (B) a minimal zero-shot baseline (task and output format only); and (C) the 14-day period-level trend prompt. The user message carries only the post texts. Full prompt texts, pipeline source, and evaluation scripts are released as an anonymous supplementary repository at <https://anonymous.4open.science/r/bd-state-annotation/>.

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